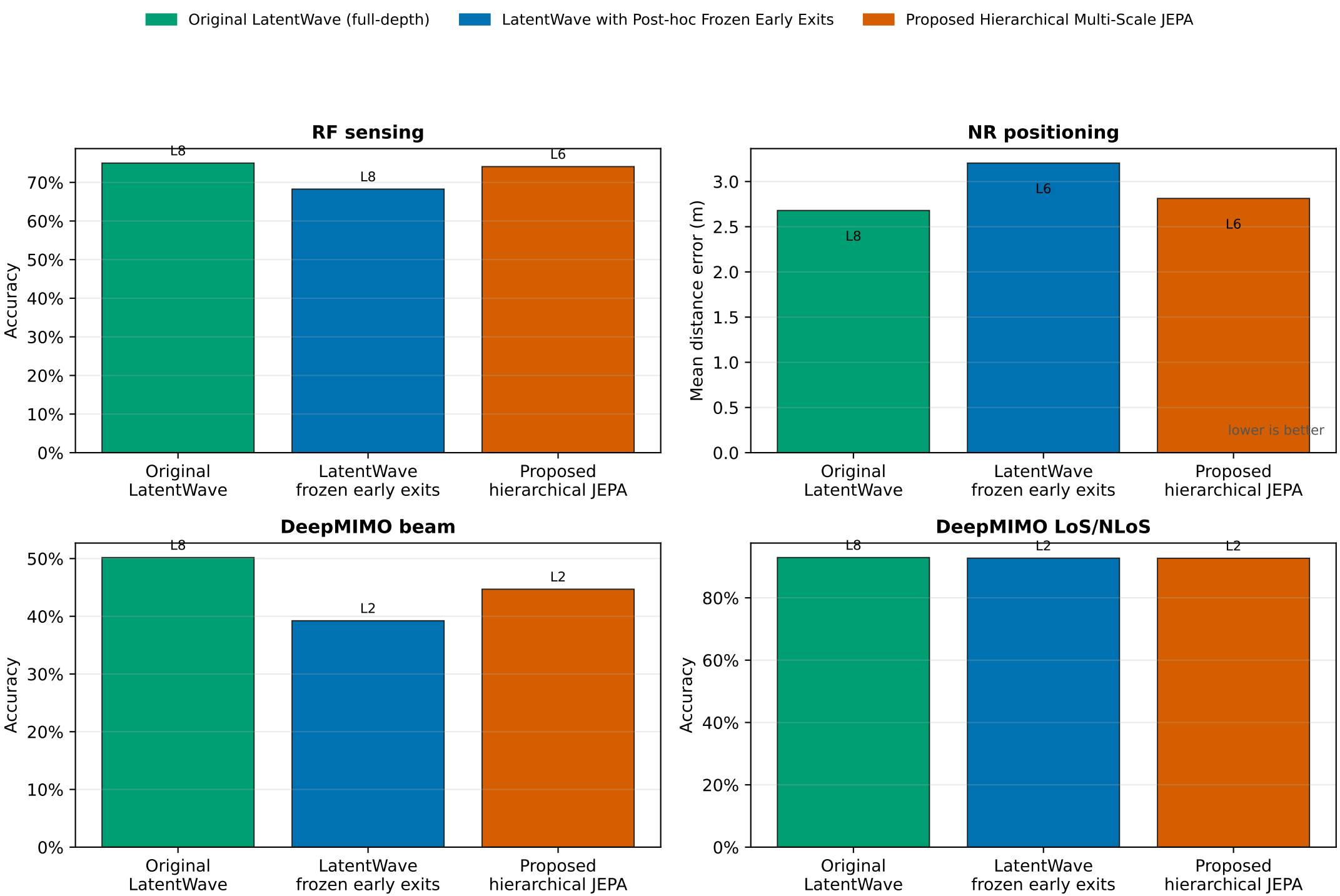
Headline downstream result at each approach's deployment choice
Original LatentWave (full-depth)
LatentWave with Post-hoc Frozen Early Exits
Proposed Hierarchical Multi-Scale JEPA
RF sensing
Accuracy
0%
10%
20%
30%
40%
50%
60%
70%
L8
L8
L6
Original LatentWave
LatentWave frozen early exits
Proposed hierarchical JEPA
NR positioning
Mean distance error (m)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
L8
L6
L6
lower is better
Original LatentWave
LatentWave frozen early exits
Proposed hierarchical JEPA
DeepMIMO beam
Accuracy
0%
10%
20%
30%
40%
50%
L8
L2
L2
Original LatentWave
LatentWave frozen early exits
Proposed hierarchical JEPA
DeepMIMO LoS/NLoS
Accuracy
0%
20%
40%
60%
80%
L8
L2
L2
Original LatentWave
LatentWave frozen early exits
Proposed hierarchical JEPA
Layer labels show full-depth L8 for Original LatentWave and the earliest exit within 5% of validation-best for the two exit methods.